



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Final Examination
Date: 3/15/2025

Total Time: 2.00 Hours.

Course Title: Numerical Analysis
Semester: Fall 2024
Total Marks: 40

Answer any four (04) of the following questions:

4 × 10 = 40

1. (a) Derive the Trapezoidal rule, Simpson's 1/3 rule. [6]
(b) Find the approximate value of $\int_0^1 \frac{1}{1+x^2} dx$ by Simpson's 1/3 rule. Find the error with the exact solution 0.7854. [4]
2. Calculate the value of $\int_4^{5.2} \ln(x) dx$ by Trapezoidal rule, Simpson's 3/8 rule, Weddle's rule. Find the error with the exact solution $[x \ln(x) - x]_4^{5.2} = 1.828$. [10]
3. (a) Solve the differential equation $\frac{dy}{dx} = 1 + xy$; $y(0) = 1$ the approximate value of y for $x = 0.4$ by Euler's method. [3]
(b) Solve the differential equation $\frac{dy}{dx} = x^2 + y^2$; $y(0) = 1$ the approximate value of y for $x = 0.3$ by second and third order Range-Kutta method and also compare the finding results. [7]
4. (a) Write the fourth-order Range-Kutta method for a second-order differential equation. [1]
(b) Solve the differential equation $\frac{dy}{dx} = x + z$; $\frac{dz}{dx} = x - y^2$; $y(0) = 2, z(0) = 1$ the approximate value of y, z for $x = 0.3$ by second and third order Range kutta method. [9]
5. (a) Write the procedure of the fixed-point method to solve a nonlinear system of equations. [3]
(b) Solve the following system of equations by fixed-point method:
$$x^2 - y^2 = 4$$
$$x^2 + y^2 = 8$$
 [7]
6. (a) Write the procedure of the Newton-Raphson's method to solve a nonlinear system of equations. [2]
(b) Solve the following system of equations by using the Newton-Raphson's method:
$$x^2 + y = 1$$
$$x + y^2 = 7$$
 [8]

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE321: Theory of Computing

Batch: 22nd

Semester: Fall 2024

Date: 17/03/2025

Final Examination

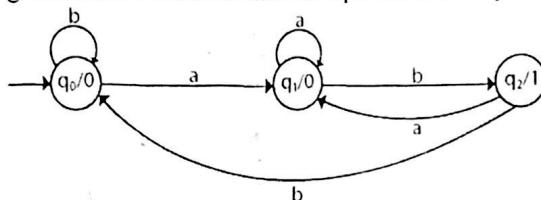
Time: 2 Hours

Total Marks: 40

Answer any **five** questions.

Marks: 5×8 = 40

- | | Marks |
|---|-------|
| 1. (a) Define Mealy machine with mentioning its possible tuples. | 2 |
| (b) Design a Mealy machine for a binary input sequence such that if it has a substring 111, the machine outputs X, if the input has substring 000, it outputs Y, otherwise it outputs Z. | 6 |
| 2. (a) Design a PDA for accepting a language $\{a^n b^{2n} \mid n \geq 1\}$. | 4 |
| (b) Design a Moore machine with the input alphabet $\{0, 1\}$ and output alphabet $\{A, B, C\}$. if it has a substring 101, the machine outputs A, if the input has substring 110, it outputs B, otherwise it outputs C. | 4 |
| 3. (a) Define Pushdown Automata (PDA). | 2 |
| (b) Derive the string "00110101" for leftmost, rightmost, and derivation tree using a CFG given by,
$S \rightarrow 0B \mid 1A$
$A \rightarrow 0 \mid 0S \mid 1AA$
$B \rightarrow 1 \mid 1S \mid 0BB$ | 2×3=6 |
| 4. (a) Write a short note on Context Free Grammar (CFG). | 2 |
| (b) If G is the grammar $S \rightarrow SbS \mid a$, show that G is ambiguous. | 6 |
| 5. (a) Differentiate between Moore and Mealy machine. | 3 |
| (b) Convert the given Moore machine into its equivalent Mealy machine. | 5 |

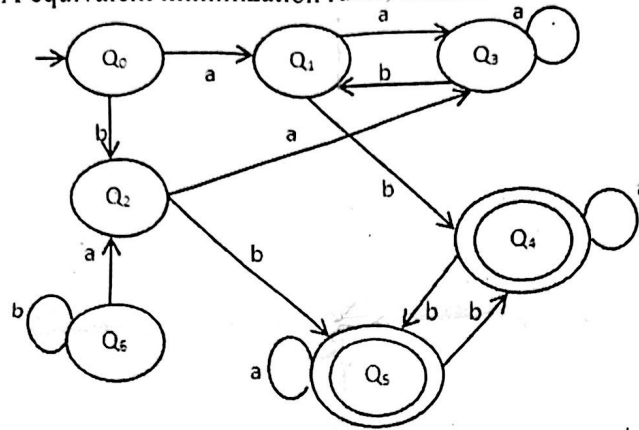


6. Suppose ,
 $S \rightarrow (L) \mid a$
 $L \rightarrow L, S \mid b \mid s$
 Find **leftmost** derivation for
- (i) $(a, (a, a))$
 (ii) $(a, ((a, a), s))$

2×4=8

7. Using the DFA equivalent minimization rules, minimize the following DFA:

8





HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE325: System Analysis and Design

Batch: 22nd

Semester: Fall 2024

Date: 12-03-2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 × 8 = 40

- ✓ (a) Describe the steps in planning the interview for data collection in system analysis. 4
- (b) List the advantages and drawbacks of using open-ended questions in data collection. 4
- ✓ (a) Illustrate the pyramid structure for interviewing. 3
- (b) The cost of developing a website over a five-month period as follows: 5

At the end of	Stage	Estimated Cost	Cumulative Estimate	Estimated Duration	Stage Completed	Actual Cost of Stage to Date	Actual Cost of Project to date
Month 1	Stage 1	\$6,000	\$6,000	1 month	100%	\$6,000	\$6,000
Month 2	Stage 2	3,000	9,000	1 month	100%	3,000	9,000
Month 3	Stage 3	3,000	12,000	1 month	100%	3,000	12,000
Month 4	Stage 4	3,000	15,000	1 month	50%	5,000	17,000
Month 5	Stage 5	3,000	18,000	1 month	0%	Not yet begun	Not yet begun

If the team is able to complete only 50 percent of stage 4 at the end of the fourth month, then calculate the revised budget Estimate at Completion (EAC).

- ✓ (a) Differentiate between logical and physical data flow diagrams. 3
- (b) What is sampling in system analysis? Explain the steps in sampling design. 5
4. (a) Define scaling in Questionnaires. Explain the problems can arise due to careless construction of scales. 4
- (b) List the guidelines to use when choosing language for questionnaires. 4
5. (a) Define data dictionary. List the components of it. 4
- (b) Assume that, the proportion of the population having the attribute $p = 0.05$, acceptable interval estimate $i = \pm 0.02$. Now, calculate the sample size if the confidence level is 90%. 4

Confidence Level (%)	Confidence Coefficient (z value)
99	2.58
98	2.33
97	2.17
96	2.05
95	1.96
90	1.65
80	1.28
50	0.67

6. (a) Briefly explain the reasons for partitioning a DFD. 3
(b) Draw a DFD for University Admission System. 5
7. Construct and explain an E-R Diagram for Library Management System. 8



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 325: Computer Peripherals & Interfacing

Batch: 22nd

Final Examination

Semester: Fall 2024

Time: 2 Hours

Date: 24/03/2025

Total Marks: 40

Answer any five questions of the followings :		5 × 8 = 40
		Marks
1.	(a) State the job of RS-232(X) protocol.	2
	(b) Write the functions of any six major pins of RS-232 port.	6
2.	(a) Why EIA/TIA-232-E standard was developed?	2
	(b) Mention the functions of Peripheral Component Interconnect bus system.	2
	(c) Draw the block diagram of a PCI bus system with proper explanation.	4
3.	(a) Why General Purpose Interface Bus is used?	2
	(b) Write the features of IEEE 4888 General Purpose Interface Bus (GPIB).	6
4.	(a) Why Universal Serial Bus was designed?	2
	(b) Write the characteristics of USB cable.	2
	(c) Shortly write on (i) Burst mode and (ii) Cycle stealing.	4
5.	(a) With proper illustration explain how CPU and DMA chip use the system bus.	4
	(b) Write the key operations of Intel 8259A Programmable Interrupt Controller.	4
6.	(a) Draw the Schematic Diagram of Memory and I/O Interfacing.	4
	(b) Explain the architecture of Intel 8255A as a Programmable Peripheral Interface.	4
7.	With necessary figure, broadly explain how Intel 8237 DMA Chip organize a complete DMA operation	8



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 323: Operating System

Batch: 22nd

Semester: Fall 2024

Date: 19/03/2025

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the following.

5 × 8 = 40

1. (a) Define deadlock and explain the four necessary conditions that must be satisfied simultaneously for a deadlock to occur. 4
- (b) Consider the data where Processes: P1, P2, P3, Resources: R1, R2
Allocations: P1 → R1 (Allocated), P2 → R2 (Allocated), P1 → R2 (Requested), P2 → R1 (Requested).
Determine whether this system is in a deadlock. Justify your answer. 4
2. (a) Define the following terms related to disk performance: 4
 - i. Seek time
 - ii. Rotational latency
 - iii. Transfer rate
 - iv. Average access time.
- (b) Explain Moving-head Disk Mechanism with figure 4
3. (a) Determine Effective Memory Access Time (EMAT) for a virtual memory with the following data: 3
Page fault=30%, Page fault service time 700ns, Memory access time=20ns
- (b) Explain Demand Paging and Page Fault with proper figure. 5
4. (a) What is a safe state in the context of deadlock avoidance? 2
- (b) Explain how deadlocks can be prevented by addressing each of the four necessary conditions. 3
- (c) Distinguish between a safe state, an unsafe state, and a deadlock state with the help of a diagram. 3
5. (a) Explain the purpose of the Banker's Algorithm and describe the data structures required for its implementation. 3
- (b) Given the following system state: Determine the safe sequence using Banker's Algorithm 5

Allocation	Max	Available
A B C	A B C	A B C
P0: 0 1 0	7 5 3	3 3 2
P1: 2 0 0	3 2 2	
P2: 3 0 2	9 0 2	
P3: 2 1 1	2 2 2	
P4: 0 0 2	4 3 3	

6. (a) Explain how segmentation works in memory management. How is it different from paging? 3
- (b) A segment table is given with the following entries: 5

Segment No.	Base Address	Limit
0	100	400
1	1200	600
2	2500	300

Translate the logical addresses (150,1), (50,0), and (275,2) into physical addresses. If an address is invalid, identify the reason.

7. (a) Explain Hardware Architecture of Paging with figure. 3
- (b) Solve the followings mathematical problems: 5
- If Logical Address is 24bits, find out the Logical Address Space.
 - A process size is 8MB, Physical Address Space is 16MB. If the Page Size is 4KB then calculate Total no of Page and Total no of Frames from the given information.
 - Virtual Address Space is 4GB and Physical Address is 32bits. If the Page Size is 23 bits then calculate Total no of Page and Total no of Frames from the given information.



HAMDARD UNIVERSITY BANGLADESH

FACULTY OF SCIENCE, ENGINEERING, AND TECHNOLOGY Department Of Computer Science and Engineering MGT 417: Industrial and Operation Management

Semester: Fall 2024

22/03/2025

Semester Final Examination

Time: 2 Hours

Total Marks: 40

Instructions:

- All parts of each question must be answered consecutively.
- Answer to questions should be neat and clean.
- Answer any Four (4) of the following questions.

No.	Question/Statement	Marks									
1.	A semiconductor manufacturing company is trying to establish a facility in between Dhaka and rangpur. Two preferred location is Pabna and Nator. The information for expansion decision is given bellow- Distance (Dhaka-Pabna 127 KM, Pabna-Nator 49 KM, Nator-Rangpur 104 KM) Sales (Dhaka 29T, Rangpur 26T) Raw materials (Dhaka 25T, Rangpur 45T) Dumping distance (Pabna 44 KM, Nator 32 KM) c) Draw a probable site map for the analysis.~ d) Based on load-distance method, which location will be preferable for the organization?	2 8									
2.	The cost schedule for two different locations for a glass manufacturing company is shown below- <table><tr><th>Location and Cost</th><th>Fixed Cost (\$)</th><th>Variable Cost (\$ per unit)</th></tr><tr><td>Munshigonj</td><td>340</td><td>4</td></tr><tr><td>Bhairab</td><td>135</td><td>5</td></tr></table> c) Calculate the breakeven point of these two locations. d) Where the company should establish the plant if the annual demand for glass is 480 unit?	Location and Cost	Fixed Cost (\$)	Variable Cost (\$ per unit)	Munshigonj	340	4	Bhairab	135	5	7 3
Location and Cost	Fixed Cost (\$)	Variable Cost (\$ per unit)									
Munshigonj	340	4									
Bhairab	135	5									
3.	c) Discuss some of the factors that can affect the location choice of a manufacturing/ service industry. d) What is plant layout? Why do you think layout analysis is important for any business firm?	5 5									
4.	c) Discuss the three demand options named as Backlogs, Backorders, and Stockout that corporate organizations generally manage to maintain the market. d) What are the planning strategies you can take to manage the human resource of your organization based on your demand options?	5 5									



HAMDARD UNIVERSITY BANGLADESH

5. PQR automobiles sources some of their leather materials from Alpha leather company. The annual demand of leather is 1000 square foot. The order and setup cost of each lot is \$200, and the holding cost per unit is \$4.
- c) Calculate the economic order quantity (EOQ) for PQR automobiles. What will be the cost for ordering the EOQ amount? 6
 - d) What will be the issue if PQR automobiles orders a lot of 87 unit? 4
6. c) Organizations always pressurize the management for lowering the inventory. Analyse the statement with proper reasoning based on different costs aspect 5
- d) Reducing supply uncertainty can help to reduce the holding of safe stock inventory. Evaluate the statement with an example. 5
-